

Grounding Lean Agents with a Curated Informal–Formal Join: a Preregistered Study with Corrective Reanalysis

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experiments and analyses executed by Claude Code agents under the author’s direction (§7)

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Preregistration: `docs/research/BRIDGE-EXPERIMENT.md`, commit `0d36f266` (2026-07-16). Versions 1 (2026-07-25) and 2 (2026-07-31) remain in git history, and the external reviews of both are preserved verbatim in `docs/research/review/`. Everything moved out of the main text lives in the companion supplement (`docs/research/BRIDGE-SUPPLEMENT.md`, typeset as `supplement.pdf`), cited below as “Supplement §Sn”. All data and code are in the private preservation repository `Deicyde/wikilean-bridge-experiment` (Supplement §S14); every number in this paper is recomputable from a named file there.

Abstract

We test whether a curated join of informal concepts and Mathlib declarations — the WikiLean Brain, served to agents as tools — improves language-model Lean performance, in a preregistered five-arm study with controls holding the same corpora accessible but unjoined. No preregistered confirmatory endpoint was graded (§3.3); everything is exploratory or corrective. A blinded audit found the hallucination oracle 13.3% precise on flags and biased against the controls; a five-rule repair validated at 90% held-out; inference moved to a commit-clustered bootstrap. Tools versus none is robust: on 100 post-Brain-index tasks Brain-arm grounded typecheck is 48% over a 21% floor (+0.27, 95% CI [+0.13, +0.40], $p=0.0004$), and formal tools collapse oracle-flagged hallucination — an upper bound on truth — to 6–11% of runs versus 30–37% (clustered $p=0.0002$). Between packages nothing favors the join: Brain minus unjoined is +0.11 ($p=0.30$), and the one decisive contrast, premise retrieval, favors formal search by 27 points ($p=2.3e-6$, exploratory). The join’s measured value is verification and routing, not content: 88% of Brain-arm hits are generated names its oracle verified, 10% graph-surfaced, and the graph holds 38% of golds. The proving phase was uninformative: the union arm sent 94% of calls to formal search; the Brain-only arm never ran. Transferable findings close the paper: instrument bias, a contamination-by-endpoint interaction, four infrastructure failure modes, and benchmark snapshot rot.

1 Introduction and contributions

A language-model agent working in Lean holds two corpora at once, informal mathematics and the formal library, and it holds them loosely: it hallucinates most at their boundary, citing declarations that do not exist for concepts it understands. Four independent 2025–26 systems (CRAMF, Aria, DRIFT, DDR; §2) converged on the same response, grounding autoformalization in concept-level retrieval, and all four report gains. But all four construct the informal→formal join per query, with learned components, and evaluate against no-retrieval or weaker-retrieval baselines. None asks the causal question: given the same corpora, does *joining* them help?

We built the join as a persistent artifact and preregistered that test. The WikiLean Brain is a curated knowledge graph whose atom — a *cell* — fuses a Mathlib declaration, a Wikidata/Wikipedia concept, and external database pages denoting the same mathematical object, with typed, provenance-carrying bonds between atoms. Agents reach it through eight MCP (Model Context Protocol) tools: `brain_bridge` (free text in, existence-verified declarations out), six graph and content tools, and `decl_exists`, a batch existence check against an index of all of Mathlib, not just the joined subset.

This paper is the corrected report of that experiment, restructured after two external reviews. Its contributions are five.

1. **The only causal-control design we could verify.** Among the 2025–26 concept-grounding systems, this is the only evaluation whose control arms hold the same informal and formal corpora accessible but unjoined (§2, §3) — the design that isolates joining itself.
2. **A robust tools-versus-none effect on grounded output** (§4). The Brain arm exceeds the no-tools floor by 27 points of grounded typecheck rate under commit-clustered inference ($p=0.0004$), the unjoined-tools arm by 16 points (clustered $p=0.012$), and formal-tool access collapses repaired-oracle-flagged citation hallucination (clustered $p=0.0002$). These survive instrument repair — including a held-out blinded revalidation — and clustering.
3. **A mechanism finding** (§5): on retrieval benchmarks the join’s measured value is *verification and routing, not content*. 88% of the Brain arm’s hits are generate-then-verify, only 10% are surfaced by the graph, and the graph contains 38% of the golds — convergent with DDR’s independent finding for existence-checked generation.
4. **Transferable evaluation-design findings** (§§3–7): a hallucination oracle whose flagged class was 13.3% precise and biased against the control arms; a contamination-by-endpoint interaction in which the apparent winner flips with the choice of endpoint; four infrastructure failure modes that silently manufacture arm differences; and benchmark snapshot rot (45% task loss in six months).
5. **The artifact itself** (§3): a characterized, provenance-carrying informal–formal join that other groups can evaluate against.

Equally plainly, **between-tool-package superiority is not established at this sample size**. The Brain-versus-unjoined contrast is +0.11 ($p=0.30$) and Brain-versus-formal-search +0.15 ($p=0.10$); the preregistered semantic endpoint was graded only by an uncalibrated LLM judge; and the Brain arm bundles the join with the verifier, so attribution between them awaits a factorial ablation (§8).

2 Related work

Concept grounding for autoformalization is no longer a novel idea; the causal test of the join is. CRAMF (arXiv:2508.06931) builds a concept-definition knowledge base from Mathlib4 — 26k formal definitions, 1k+ concepts, assembled by an unsupervised LLM pipeline — and injects retrieved definitions into a single translation call. Aria (arXiv:2510.04520, ICLR 2026) decomposes each informal statement into a concept dependency graph, grounds each node against Mathlib on the fly, and grades outputs with a semantic checker (AriaScorer). DRIFT (arXiv:2510.10815, ICLR 2026) decomposes statements into concept-level sub-queries for a Mathlib-fine-tuned premise retriever and grades with BEq+. DDR (arXiv:2511.11990) has a model *generate* candidate dependency names and validates each against the library with a suffix-array existence check. DDR is the closest published analog of `decl_exists`, and independent convergence on existence verification of generated names as the anti-hallucination mechanism (§5).

Three adjacent lines complete the picture. TheoremGraph (arXiv:2606.25363) builds statement-level graphs on both registers and bridges them by embeddings; it is a dataset, not agent tooling. LeanSearch-v2 (arXiv:2605.13137) is learned declaration-level premise retrieval over a formal-only corpus with a fixed downstream prover loop. ProofNetVerif/BEq+ (arXiv:2406.07222) anchors the semantic-grader axis and is the ready-made instrument for the equivalence leg this study left ungraded; on that axis Aria and DRIFT are ahead of us.

Table 1: The 2025–26 concept-grounding landscape on the reviewer’s axes. “No-join control” = a control holding the same corpora accessible but unjoined.

System	Unit of retrieval	Join	Coverage	Side	Downstream eval	No-join control
CRAMF (2508.06931)	concept→definition	learned (LLM pipeline)	26k defs / 1k+ concepts	system (RAG)	stunt autoformalization	no (no-RAG baseline)
Aria (2510.04520)	concept node, per-stmt graph	learned, ephemeral per query	Mathlib (on the fly)	agent	stunt autoformalization	no
DRIFT (2510.10815)	concept sub-query→premise	learned (fine-tuned DPR)	Mathlib	system (pipeline)	stunt autoformalization	no (weaker retrieval)
DDR (2511.11990)	generated decl name	learned + existence check	full library (suffix array)	system	stunt autoformalization	no
TheoremGraph (2606.25363)	statement	learned (embed + LLM judge)	38k decls, 25 projects	neither (dataset)	retrieval study only	no
LeanSearch-v2 (2605.13137)	declaration (premise)	learned (embed+verank), formal-only	all Mathlib	system (search API)	fixed prover loop	no-tools, not unjoined
ProofNetVerif (2406.07222)	n/a (evaluation)	n/a	n/a	n/a	metric validation	n/a
Wikibrain (this study)	cell (concept ≡ decl ≡ article ≡ DB page)	curated, human+AI-moderated, persistent	7,430 decls joined; existence index = all of Mathlib	agent (MCP tools)	typecheck, retrieval, kernel proving	yes — unjoined-corpora arms

Semantic graders: CRAMF translation metrics; Aria AriaScorer; DRIFT BEq+; DDR accuracy/stability; TheoremGraph LLM judge on matches; LeanSearch-v2 none; ProofNetVerif BEq+ + 3,752 labels; Wikibrain an exploratory uncalibrated judge (a gap).

The table encodes one positioning fact. Every 2025–26 concept-grounding system joins per-query with learned components and lacks an unjoined control; the Brain occupies the persistent-curated-database quadrant, and this study’s D-versus-C/E contrast is, as far as we could verify, the only causal test of the join itself.

3 The Brain artifact and experimental design

This section characterizes the artifact under test, then the preregistered design, its deviations, and what we know about execution fidelity and holdout status.

3.1 The system under test

Table 2 characterizes the Brain under test, **post hoc, from the 2026-08-01 build** (recompute: `bench/analysis/brain_artifact.py`). No experiment phase queried this exact build, and the per-phase build commits were not recorded — a bookkeeping gap. What we can bound is this: the fresh-set holdout was checked against the 2026-07-03 snapshot universe, and the one documented mid-campaign change to the graph is the 2026-07-18 verified discovery fold of §3.5, which absorbed a single fresh gold (fresh_025). The table therefore characterizes the artifact family the arms queried, not any single run’s view of it.

Table 2: The Brain, 2026-08-01 build (Mathlib pins: 2026-07-04 harvest + 2026-07-18 discovery folds).

Quantity	Value
Cells (atoms)	20,880 (4,095 multi-organ; largest cell 28 organs)
Organs	29,051 — decl 19,611 · concept 2,969 · article 728 · external-DB page 3,467 · arXiv statement 2,276
Mathlib declarations joined	7,439 distinct = 2.34% of 317,257 doc-gen4 Mathlib names; 1,197 share a cell with ≥ 1 concept
Concepts	2,969 distinct QIDs in cells (+99 field concepts held at folder altitude)
Synapses	100,797 aggregated edges carrying 132,708 bonds — depends 85,652 · mentions 12,361 · links 11,597 · invocation 9,204 · co-page 7,304 · cites 3,532 · other 3,058
Provenance, organ attachments	human 5.5% · ai-moderated 30.9% · ai 0.1% · automated 63.6%
Provenance, synapse traces	human 0.3% · ai-moderated 1.8% · automated 98.0%
External DBs (cell-anchored pages)	MathWorld 1,360 · nLab 564 · ProofWiki 426 · EoM 409 · Stacks 268 · PlanetMath 250 · 5 others
Benchmark-target coverage	MathlibQR golds 65/171 (38.0%) · MathlibMPR premises 47/341 (13.8%) · fresh-100 golds 1/100

Construction is a deterministic, no-LLM build (external-DB ingest + Wikidata/Mathlib harvests $\rightarrow$ organ layer $\rightarrow$ cell merge $\rightarrow$ synapse aggregation); AI-generated joins enter only through verified proposal folding, and acceptance suites gate publication. The coverage row matters twice below: the Brain is a *map* of Mathlib (2.34% of names joined, existence index over all of them), and it contains the gold answer for only 38% of the concept-retrieval benchmark (§5).

3.2 Preregistered design

The preregistered hypotheses were that an agent holding the joined dictionary beats one given informal and formal search separately (P1), at no higher cost (P2), with better grounding (P3). Five arms ran with identical model, prompt, and budgets within each phase; only the tool manifest differs:

Arm	Tools	Isolates
A <code>no_tools</code>	none	floor / memorization
B <code>informal</code>	Wikipedia + nLab search/fetch	informal reasoning alone
C <code>formal</code>	loogle + ripgrep + source read (Mathlib checkout)	the LeanSearch-class status quo
D <code>wikibrain</code>	the Brain MCP — all eight tools	the join + the existence verifier (not separable here)
E <code>B+C_unjoined</code>	B’s and C’s tools, no join	the unjoined-corpora control

Two design facts matter up front. First, D alone holds `decl_exists`, and hallucination-free citation is a conjunct of the primary metric — so D versus E measures the *bundled Wikibrain package*, not the join (715 `decl_exists` attempts on the 100 fresh tasks; 682 reached the tool, 681 succeeded — the other 33 called a nonexistent bare tool name). Second, E failed its manipulation check: 4 informal-tool touches across its 100 fresh runs against B’s 345 — it behaved as a formal-search agent with a larger manifest.

Tier 1 ran `claude-haiku-4-5` on ProofNet# (371 tasks; 341-task eval split; pinned fork of arXiv:2406.07222) and on a 100-task **fresh set** drawn from theorems merged into Mathlib master 2026-07-03 $\rightarrow$ 07-16, graded by REPL typecheck on pinned toolchains.

What a fresh task is. Candidates were every theorem/lemma line added to Mathlib master between the Brain snapshot (2026-07-03) and the 2026-07-16 head `9944fe29` — 1,326 added lines across 44 commits, pre-filtered to the 187 carrying an author-written docstring, of which 100 were kept (bespoke helpers and statements with unrecoverable hypotheses dropped). Each task’s gold is the exact Mathlib master statement (`:= sorry`), elaboration-checked on the fresh pin; the agent sees only the informal statement and must produce a Lean statement of the same theorem. The informal statements are identifier-stripped natural-language paraphrases of the golds’ docstrings, written by a Claude Opus 4.8 agent session. That session shares a vendor with the subject models, and docstring-derived NL for formula lemmas is inherently close to the Lean: both are validity caveats we cannot remove. Held-out guarantee: every kept declaration is absent from both the pinned 388k-declaration TheoremGraph universe and the Brain’s node set. One honest gap remains — the construction script was never committed, so the docstring→NL step is documented by the commit record and `fresh_tasks.stats.json`, not by re-runnable code. Determinacy was screened post hoc by two independent AI annotators (86 and 83 of 100 judged determinate; 74 by both — the primary subset, Supplement §S1).

The exploratory second phase (“v2”, `claude-sonnet-5`) ran retrieval arms N (no tools), F (formal search), W (the Brain), WF (union + a tool manual), and U (bare union) on third-party-graded benchmarks (§5), and a one-shot proving protocol on SorryDB (§6). Each phase used one model and one seed per (task, arm) pair.

3.3 Deviations from preregistration

The complete 25-component execution inventory is Supplement §S1; six deviations change interpretation:

#	Deviation	Consequence
1	Primary endpoint faithful@budget (BEq+ equivalence) never graded in campaign; the grader was a stub	zero confirmatory results; a post-review blind LLM-judge pass over all 500 fresh outputs is the exploratory substitute (§4.2)
2	Second model class on the primary set not run	Tier-1 all-Haiku, v2 all-Sonnet; capability-band generality unknown
3	Three reseeds + pass@k not run	one seed everywhere; no run-to-run variance estimate
4	tokens-to-solve never computed	P2 (the cost half of the success criterion) untested
5	30-turn budget advisory only (stated in prompt, no CLI cap)	overruns on the repaired rows C 50 / D 38 / E 48 of 100, max 88 turns (the as-run count for E was 32 — its 31 outage rows died at turns=1 and masked overruns); sensitivity in Supplement §S5
6	Arm-E fresh block: 31 contiguous infrastructure-failed rows	repaired by rerun (§3.4); outage-basis tables in Supplement §S2

3.4 Execution fidelity

Four infrastructure defects were caught by cross-checking transcripts against results; none was visible in topline files.

(1) *The 429 tail.* Rows fresh_069–099 — 31 contiguous tasks at the tail of E’s sequential block — died with session-limit 429 errors and were rerun on 2026-07-27 with E’s exact code path against a read-only archive of the same tree (commit `19a90209`).

(2) *The MCP non-blocking attach leak.* The agent CLI attaches its MCP servers without blocking the first model turn, so a run can “complete” having silently never had tools; the repair driver detects this from the stream-json init event and retries.

(3) *The cold-start race* — found and repaired. Under concurrent batch cold starts the same non-blocking attach silently detooled whole waves: 13/49 rows of the first U launch and 175/810 zero-tool rows of the original QR-810 F grid carried the pending-at-init signature (12/12 sampled confirmed). The v2 runner now condemns such rows and staggers its first wave (commit `834a130a`), and all 194 affected rows (F: 175 QR + 15 MPR; W: 2 + 2 — W’s Brain server is a fast-attaching local worker, so the race fell almost entirely on F) were rerun attach-verified under the fixed harness, the originals byte-preserved in `bench/v2/runs/agent/race_condemned_archive/`. The final §5 grid audits **zero race rows** in every arm × benchmark (commits `99a2075f`, `f041928a`, `dd3eb689`). The bias had run *against* F — race rows behaved as no-tools — that is, against our own tooling conclusion.

(4) *The REPL silent fallback.* The typechecker silently fell back from the persistent REPL server to a ~60s single-shot Lean boot on the Tier-1a pin, tainting a scoring pass with timeout failures until it was caught and rerouted with honest labels (commit `2a9f6b91`).

Could the Tier-1 headline grids have run toolless? The Tier-1 rows predate the init-signature capture and record no attach event, so the audit is indirect: nonzero tool calls prove attachment, and a zero-tool row is the worst case. On the fresh set the tooled Lean arms show C 0 / D 1 / E 1 zero-tool rows (each a single-turn, ~ 85 s row — the plausible silent-detoiling suspects, $\leq 1\%$ per arm), and on eval-341 C 0 / D 0 / E 4. Arm B’s 23 fresh zero-tool rows are single-turn answers consistent with declining a rarely useful wiki toolkit rather than detoiling; arm D additionally preflighted its MCP server and aborted on failure. The Tier-1 grids are thus not attach-verified row-by-row as the v2 grid is, but worst-case contamination is one row per tooled Lean arm on the fresh set (`v3_gate_fixes.json`).

3.5 Fresh-set isolation status

The fresh set is **post-Brain-index**: held out from the Brain’s declaration universe and node set — verified per-gold, with one leak. `fresh_025`’s gold entered the Brain through a later (2026-07-18) verified discovery fold, a second Mathlib-entry channel the holdout check did not cover; the strict claim is **99/100**. The set was *not* held out from the formal-search arms’ sources: 51/100 golds are exposed in the checkout C and E read (own-module basis; full strata Supplement §S4). Exposure interacts with the choice of endpoint, and §4.2 measures that interaction directly.

Terminology used below. A **verified rename** is a dead $\rightarrow$ current name pair (e.g. `Basis` $\rightarrow$ `Module.Basis`) from our rename ledger, each pair checked against the current declaration index; the union oracle accepts either side. A `decl_exists` attempt is **successful** when the call returns a verdict, not when the name exists. The judge’s **strict** grade requires the same proposition under the same hypotheses; **evaluated** requires mathematical equivalence at high confidence. **fair-810** is the LSv2-released MathlibQR subset whose 171 gold declarations lie in the universe shared by all compared systems, times up to six paraphrase styles per gold = 810 queries. **R@10** (recall at ten) scores whether a gold declaration appears in a ranked list’s top ten; **nDCG@10** (normalized discounted cumulative gain) additionally rewards ranking it early. **group-recall@10** (MPR): each task’s gold is a set of premise groups of interchangeable names; the score is the fraction of groups with ≥ 1 member in the top 10. An **RD** is a risk difference between two arms’ rates; a **Wilson 95% CI** is the binomial confidence interval we quote for single proportions; an exact **McNemar** test compares paired arms on their discordant tasks. A **sign count** such as 34/35 counts gold-declaration clusters where one arm’s mean beats the other’s (ties dropped). **Own-module** exposure counts a gold as exposed when its declaration name appears as a header in the task’s own module file in the pinned checkout.

4 Statement formalization results (claude-haiku-4-5, fresh-100)

4.1 Grounded typecheck under the repaired instrument

The register label comes first: the preregistered primary endpoint (`faithful@budget`) was never graded (§3.3, deviation 1), so grounded typecheck is a post-hoc component of it under a post-hoc analysis plan — corrective-exploratory, not confirmatory.

The primary metric is the **grounded typecheck rate**: the run produced a declaration, cited zero names absent from a union oracle (`doc-gen4` declaration data $\cup$ verified renames), and the declaration typechecked on the pinned toolchain. It does not establish that the statement formalizes the prompt — that is §4.2’s question. The citation surface is the run’s final Lean block (`output_lean`, the last fenced Lean block of the final answer — nothing else in the transcript is scanned), swept by a capitalized-identifier heuristic with no Lean parser, so it also picks up names in comments and docstrings inside that block; the oracle then matches exact fully-qualified strings. That is why a typechecking run can still carry a flagged citation: a name in a comment is never elaborated, and an idiomatic namespace-short name under **open** elaborates while failing exact-string lookup — the repair’s main business, below. §4.3’s run-level rates inherit the same surface.

The instrument had to be repaired first. A blinded audit of the hallucination oracle (60 distinct cited names, seeded and stratified, graded against raw `git grep` evidence with verdicts sealed) found the flagged class only **13.3% precise** [5.3, 29.7]: most flags were real declarations cited by their conventional namespace-short names, plus extractor noise. Five mechanical repair rules (drop comment, import, and self-declaration tokens and single-letter dot-notation heads; resolve single-segment namespace prefixes) agree with the blinded truth on 59/60. The rules were formulated after that audit unsealed, so 59/60 is an in-sample fit; a second, disjoint 40-name blinded sample (seed 20260802) measured the repair out of sample: binary

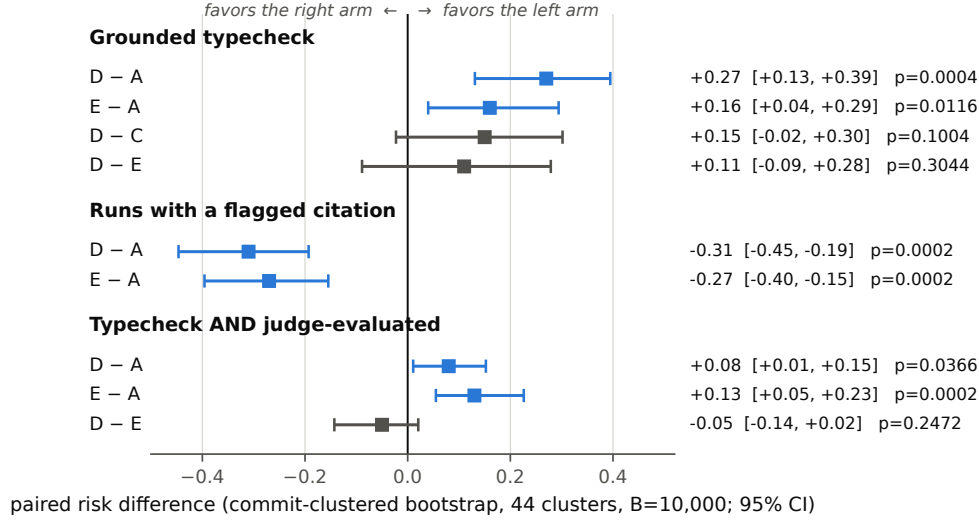


Figure 1: Commit-clustered paired risk differences with 95% CIs (fresh-100, repaired instrument; 44 commit clusters, B=10,000) for the headline contrasts, grouped by outcome. Grounded typecheck and the flagged-citation rates are corrective-exploratory (the preregistered endpoint was never graded, §3.3); the conjunction adds the uncalibrated-judge leg and is exploratory only (§4.2). Flagged-citation RDs are negative because fewer flags are better; flagged rates are upper bounds on true hallucination (§4.1). Blue intervals exclude zero. Sources: `success_repaired.json`, `v3_gate_fixes.json`.

agreement $36/40 = 90\%$ [77, 96], recall on true fabrications $3/3$, but flagged-class precision only 43–57%. The raw oracle’s invalidity replicates almost exactly (15% precision held-out), the repair transfers most of its value, and a repaired flag remains closer to a coin toss than a confirmation — so repaired hallucination rates are **upper bounds** on true rates. The residual false-flag modes are namespace-style-correlated and land on the free-text tool arms, so they cannot manufacture D’s advantage (protocol and error census: Supplement §S3). The original bias was directional — false flags hit the control arms hardest — and repairing it moves C by +8 successes, E +7, D +6, A +1, B 0 (per-row detail Supplement §S3). All headline numbers below use the repaired instrument; raw-instrument tables, the outage bases (errors-as-failures and completed-69), and the earlier Wald/McNemar mismatch are Supplement §S2.

arm	grounded typecheck (repaired)	Wilson 95% CI
A no tools	21/100 = 21.0%	[14.2, 30.0]
B informal	22/100 = 22.0%	[15.0, 31.1]
C formal	33/100 = 33.0%	[24.6, 42.7]
D wikibrain	48/100 = 48.0%	[38.5, 57.7]
E B+C unjoined	37/100 = 37.0%	[28.2, 46.8]

The 100 tasks are not independent draws: they come from **44 source commits** and 57 files, with conspicuous sibling families (a 9-task sum–integral family, an 8-task bounded-variation file). Our single main-text inferential framework is therefore the **commit-clustered paired bootstrap** (44 clusters, B=10,000; interval and p-value read off the same resampling distribution):

pair	RD	95% CI	p
D – A	+0.27	[+0.131, +0.395]	0.0004
D – C	+0.15	[−0.023, +0.302]	0.100
D – E	+0.11	[−0.089, +0.279]	0.304

E versus A is also robust under the same framework: +0.16 [+0.04, +0.29], clustered $p=0.012$ (exact McNemar 24/8, $p=0.007$). The same clustered framework was run on every A-floor and D/E/C contrast reported as significant in §§4.2–4.3 below, and all survive it (`v3_gate_fixes.json`); the remaining p-values there are unclustered McNemars, labeled as such — anti-conservative for the two significant B-comparisons, immaterial for the nulls. The picture is plain: **tools-versus-none is solid, while between tool packages**

the matched comparisons are inconclusive at this sample size. Clustering is not pedantry here — on the raw instrument, 83% of D’s net paired advantage over E sits in 2 of the 44 commits, exactly the pseudoreplication the clustered bootstrap absorbs. On the contaminated ProofNet# eval split the arms sit within a few points of each other at 58–64% with the no-tools arm at 59.8% — consistent with benchmark familiarity, though the fall to 21% on fresh tasks changes benchmark, difficulty, domain, and recency at once and is not a clean memorization quantification. The eval split was not re-graded under the repaired oracle (280 affected rows; §7).

4.2 Semantic faithfulness (exploratory; uncalibrated judge)

The preregistered equivalence endpoint went ungraded in the campaign, so after review 1 we ran a blind LLM-judge pass over all 500 fresh outputs (judge `claude-sonnet-5`; sees the informal statement, the gold with its binders, and the candidate; no arm identity, no tools; 0 errors; self-consistency on a fixed 50-item re-grade: strict 98%, evaluated 100%). **The judge is uncalibrated — the preregistered 50-item human calibration remains undone — so nothing in this subsection is confirmatory.** The story has three layers.

Layer 1 — the inversion. On judge-evaluated equivalence (mathematical equivalence at high confidence), the five arms score: A 17.0% [10.9, 25.6] · B 16.0% [10.1, 24.4] · C 51.0% [41.3, 60.6] · D 19.0% [12.5, 27.8] · E 53.0% [43.3, 62.5]. D sits at the no-tools floor (D-vs-A McNemar 7/5, $p=0.77$) while C and E sit far above it (D-vs-E clustered -0.34 [$-0.46, -0.22$], $p=0.0002$). Taken alone this says the unjoined arms produce far more faithful statements — and that the Brain arm does not beat no tools at all on this endpoint.

Layer 2 — the exposure explanation. C and E hold source grep over a checkout containing 51 of the 100 golds (§3.5), and on those tasks they can simply transcribe: their evaluated-equivalence rates are .706 and .686 on exposed tasks against .306 and .367 on unexposed ones, while D is flat (.196/.184). The inversion concentrates exactly where the gold was retrievable; it narrows sharply, but does not vanish, on the unexposed half (Supplement §S11). The unexposed residual (C/E at .31/.37 against D’s $\sim .18$) is not explained by exposure, and we cannot exclude **verifier steering**: the generate-then-verify loop may pull D toward statements that are existence-verified and typecheckable but semantically weaker — groundedness traded against faithfulness. The human grading queue below is where that hypothesis gets adjudicated.

Layer 3 — the conjunction. On grounded-typecheck $\wedge$ judge-evaluated — the closest available analogue of the preregistered faithful@budget — the arms return to parity. Under the repaired typecheck leg (`conjunction_repaired.py`; the judge ran before the oracle repair, so Supplement §S11 reports both instruments): C 20, D 16, E 21 per 100 (raw leg: 16/15/18). Every between-tool contrast is null (D-vs-E clustered $p=0.25$; D-vs-C $p=0.50$ and E-vs-C $p=1.0$, unclustered) and the tools-versus-none contrasts strengthen — E-vs-A $+0.13$, clustered $p=0.0002$; E-vs-B $p=0.004$ (unclustered); and D-vs-A crosses to $+0.08$, clustered $p=0.037$. The instrument repair moves no conclusion here: parity between tool packages, tools above the floor.

The transferable warning is the interaction itself: with contaminated items in a task set, the apparent winner *flips with the endpoint* — D leads on grounded typecheck, C/E lead on judged equivalence by transcribing retrievable golds, and the conjunction returns parity. The 36 D/E-discordant tasks are queued for blinded human grading as the first slice of the still-owed calibration.

4.3 Hallucinated citations

The table below counts runs with ≥ 1 repaired-oracle-flagged citation — an upper bound on true hallucination, per the held-out precision in §4.1 — at $n=100$ per arm:

arm	runs with ≥ 1 hallucination	Wilson 95% CI
A no tools	37/100	[28.2, 46.8]
B informal	30/100	[21.9, 39.6]
C formal	11/100	[6.3, 18.6]
D wikibrain	6/100	[2.8, 12.5]
E B+C unjoined	10/100	[5.5, 17.4]

Tools-versus-none survives everything: D-vs-A -0.31 [$-0.45, -0.19$], clustered $p=0.0002$ (E-vs-A -0.27 , clustered $p=0.0002$; D-vs-B $p=3.9e-5$, unclustered). The between-tool contrasts dissolve: D-vs-E clustered

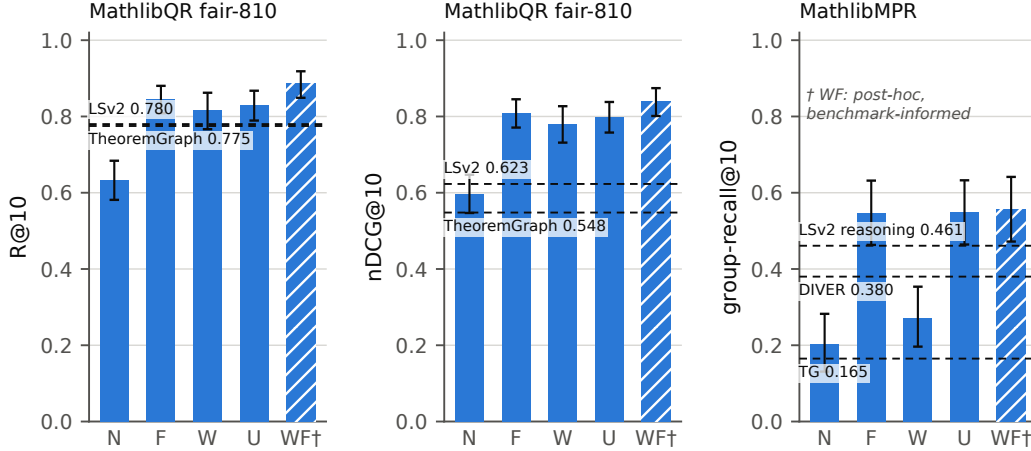


Figure 2: The repaired-grid five-arm retrieval results (exploratory) with declaration-clustered (QR) and task-bootstrap (MPR) 95% CIs, against the published anchors (dashed). WF (hatched) is post-hoc and benchmark-informed (§5, Supplement §S8); published anchors are single-call systems (MPR: LSv2’s reasoning mode is itself iterative) — context, not controlled comparison. Source: `grid_repaired.json`, `union_ablation.json`.

$p=0.44$, D-vs-C clustered $p=0.31$ — direction preserved, but $n=100$ cannot distinguish them. The raw oracle had shown D roughly $3\times$ better at citation level; most of that gap was **citation style, not grounding**. D copies fully-qualified names out of tool payloads (10 short-name reclassifications under repair) while C and E write idiomatic namespace-short names under `open` (65 and 71 reclassifications), which an exact-string oracle falsely flags. All four fabricated names confirmed in the blinded sample came from the arms without formal tools (A/B); none from C, D, or E — and the held-out sample replicates this: all three of its confirmed fabrications sit in A/B as well (Supplement §S3).

5 Retrieval scope and ablations (claude-sonnet-5, exploratory)

The v2 phase uses graders that predate the arms: third-party expert gold labels from the LeanSearch-v2 release (CC BY 4.0). MathlibQR fair-810 poses 810 paraphrase-style queries over **171 distinct gold declarations**, so all inference is declaration-clustered; MathlibMPR poses 69 post-cutoff premise-retrieval tasks, one per PR. All agent rows below are the **race-repaired grid** (§3.4): the 194 condemned F/W rows were rerun attach-verified, and the final grid audits race-free (`bench/analysis/grid_repaired.py`). **WF is a post-hoc, benchmark-informed condition**: its prepended manual distills measurements from the same evaluation queries it was then scored on (manual verbatim and commit timeline: Supplement §S8). Published rows are the numbers the respective papers report for their own systems — single-call retrievers on the QR anchors; on MPR, LSv2’s reasoning mode is itself an iterative sketch-retrieve-reflect pipeline, not a single call — context, not controlled comparison. (Anchor provenance: 0.775, 0.780, 0.623, and 0.461 are verified in `related_work_notes.md`; TheoremGraph nDCG 0.548, TheoremGraph MPR 0.165, and DIVER 0.380 are recorded from the papers’ tables in the pre-run design notes, `docs/research/BRIDGE-V2-BENCHMARKS.md`.)

Table 3: MathlibQR fair-810 — concept retrieval (repaired grid).

system	R@10	decl-clust. 95% CI	nDCG@10	95% CI	\$/query	calls/query
published: TheoremGraph	0.775	—	0.548	—	—	1
published: LSv2 retriever+reranker	0.780	—	0.623	—	—	1
system-mode <code>brain_bridge</code> (one call, no LLM)	0.036	—	0.031	—	~0	1
agent N (no tools)	0.633	[0.581, 0.684]	0.598	[0.547, 0.647]	0.08	0
agent F (formal)	0.846	[0.808, 0.880]	0.809	[0.771, 0.845]	0.14	2.8
agent W (Brain)	0.816	[0.767, 0.862]	0.781	[0.731, 0.827]	0.20	3.5
agent U (bare union, no manual)	0.830	[0.789, 0.868]	0.799	[0.758, 0.838]	0.20	3.1
agent WF (union + manual; post-hoc)	0.885	[0.849, 0.919]	0.839	[0.801, 0.874]	0.21	4.2

Declaration-clustered paired contrasts on the repaired grid: **F – W is a null** ($R@10 +0.030$ $[-0.015, +0.076]$, Wilcoxon $p=0.086$, sign 34/35) with different textures — W wins the special-case paraphrase style (nDCG 0.522 vs 0.356), F the Lean-syntax styles. **WF – F is $+0.040$ $[+0.015, +0.064]$ ($p=0.0097$)** and

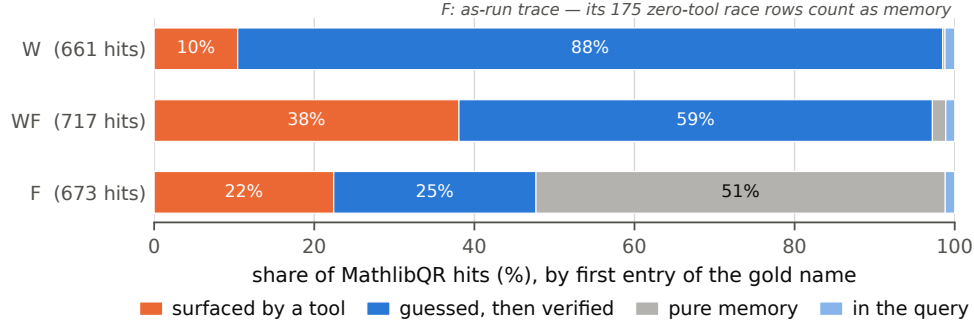


Figure 3: Provenance decomposition of MathlibQR hits (exploratory): where the gold name first entered the transcript. “Guessed, then verified” means the model wrote the name and `decl_exists` (or another tool) confirmed it; the residual sliver is names present in the query itself. The trace pass is as-run — F’s 175 zero-tool race rows count as memory, an upper bound (§5). Source: `retrieval_provenance.json`.

WF — W +0.069 [+0.031, +0.108] (p=0.0003), but WF carries the post-hoc label — and the U ablation now decomposes it: U, holding the identical W $\cup$ F toolset with no manual, is indistinguishable from F (U — F -0.016 [-0.039, +0.005], p=0.13), while WF — U is +0.056 [+0.032, +0.080] (p=1.3e-4). Note also that repaired F alone already exceeds both published QR anchors (R@10 0.846 vs 0.780/0.775) — single-call systems against a multi-call agent, so context, not a controlled comparison. The system-mode row is the API deficiency in one number: a single `brain_bridge` call scores 0.036 where an agent iterating over the same API reaches 0.816 — the free-text entry point is a label/alias resolver, not a semantic retriever.

N’s format failures explain little. 143/810 N rows returned no ranked list. A maximally lenient extraction pass applied identically to every arm closes only 5.9–8.1% of N’s deficit (0.633 $\rightarrow$ 0.648), and even the oracle ceiling — every empty N row scored as a rank-1 hit — leaves N below every toolset arm. N’s empty rows are dominated by hallucinated tool calls, not wrapper noncompliance (Supplement §S6).

Where the answers actually came from. For every QR hit we traced the chronologically first entry of the gold name into the transcript (full-transcript resolved pass):

arm	hits	surfaced by a tool	guessed, then verified	pure memory
W	661	69 (10.4%)	582 (88.0%)	2 (0.3%)
WF	717	273 (38.1%)	424 (59.1%)	12 (1.7%)
F	673	151 (22.4%)	170 (25.3%)	344 (51.1%)

(The remaining $\sim$ 1% appear in the query itself. This trace pass predates the race repair: F’s row includes its 175 zero-tool race rows, whose hits count as “pure memory”, so F’s memory share is an as-run upper bound; W and WF are materially unaffected — 2 and 0 race rows.) W’s score is not the graph surfacing answers: it surfaced 10.4% of hits, and it could not have surfaced most — the Brain holds only 65/171 (38%) of the QR golds. Instead the model generates candidates and `decl_exists` confirms them, and the loop genuinely filters: in 81% of W’s verified hits the oracle also *rejected* at least one candidate along the way. **On these benchmarks the join’s measured value is verification and routing, not content** — convergent with DDR’s finding that existence-checked generation beats selection-RAG. WF’s higher surfaced share (38.1%) is the routing half: it greps and loogles when holding syntax, bridges when holding words.

MathlibMPR — premise retrieval (group-recall@10, task-bootstrap CIs): N 0.203 [0.131, 0.282] · W 0.272 [0.196, 0.354] · F 0.547 [0.463, 0.632] · U 0.549 [0.464, 0.633] · WF 0.557 [0.472, 0.642]; published anchors LSv2 reasoning-mode 0.461, DIVER 0.380, TheoremGraph 0.165; system-mode Brain 0.000. A generic Sonnet agent with grep and loogle *exceeds* the specialist system’s reasoning mode (0.547 vs 0.461) — the as-run 0.453 had been deflated by F’s 15 race rows. W trails F by 27 points (F — W +0.275 [+0.185, +0.369], p=2.3e-6) — the preregistered concept $\neq$ premise boundary measured on our own tools, and the one decisive between-package retrieval contrast, in formal search’s favor. The as-run grid’s marginal WF — F advantage (+0.104, Wilcoxon p=0.049) was a race artifact: repaired, WF — F is +0.010 [-0.068, +0.088] (p=0.73) and WF — U +0.009 (p=0.82) — neither the manual nor the union adds anything on MPR.

The bare-union ablation and the final contrast set. U holds the identical $W \cup F$ union toolset as WF with no manual, run entirely under the condemn-and-retry protocol with a staggered first wave (879 attach-verified rows, 0 errors; its first launch is what surfaced the cold-start race, §3.4). The repaired-grid contrasts (declaration-clustered on QR, task-paired on MPR; `grid_repaired.py`, seed 20260727, B=10,000):

contrast	metric	diff	95% CI	excl. 0	Wilcoxon p
WF – F	QR R@10	+0.0395	[+0.0148, +0.0636]	yes	9.7e-3
WF – F	QR nDCG@10	+0.0305	[+0.0099, +0.0515]	yes	6.4e-3
WF – F	MPR gR@10	+0.0101	[−0.0679, +0.0882]	no	0.73
WF – U	QR R@10	+0.0556	[+0.0320, +0.0802]	yes	1.3e-4
WF – U	QR nDCG@10	+0.0402	[+0.0187, +0.0628]	yes	1.2e-3
WF – U	MPR gR@10	+0.0085	[−0.0664, +0.0833]	no	0.82
U – F	QR R@10	−0.0161	[−0.0385, +0.0050]	no	0.13
U – F	QR nDCG@10	−0.0097	[−0.0294, +0.0095]	no	0.50
U – F	MPR gR@10	+0.0017	[−0.0725, +0.0713]	no	0.83
F – W	QR R@10	+0.0297	[−0.0150, +0.0756]	no	0.086
F – W	QR nDCG@10	+0.0283	[−0.0110, +0.0684]	no	0.28
F – W	MPR gR@10	+0.2747	[+0.1851, +0.3688]	yes	2.3e-6

The decomposition is clean. On concept retrieval the active ingredient is the manual — a test-set-tuned upper bound (Supplement §S8), not a portable prescription — while on premise retrieval formal tools are the entire story and the bare union is inert everywhere (U – F null on all three metrics). Repaired F alone already exceeds the published anchors on both benchmarks (QR R@10 0.846 vs 0.780; MPR 0.547 vs 0.461).

6 Downstream proving: SorryDB (claude-sonnet-5, exploratory)

SorryDB (arXiv:2603.02668) serves unsolved `sorrys` from live Lean repositories; success is the repository *building* with your proof — kernel-graded, nothing to memorize. Our refrozen split holds 171 tasks across 10 repositories; the protocol is one-shot and build-free (goal state + a ± 60 -line window; explicit honest-abstention rule), and every one of the 203 candidate proofs has a definitive verdict — 183 kernel pass/fail, plus 19 unspliceable rows and 1 verify-timeout that never reached the kernel. (Denominators reconciled: 171 frozen tasks; run rows N 168 / F 169 / WF 171 — 3 N and 2 F runner losses; candidates N 58 + F 71 + WF 74 = 203.) Results: N 2/171 = 1.2% [0.3, 4.2] · F 9/171 = 5.3% [2.8, 9.7] · WF 10/171 = 5.8% [3.2, 10.4] (Wilson); repo-clustered bootstrap CIs are [0.0, 2.7] / [0.0, 10.7] / [0.0, 12.2] — F’s and WF’s widen, while N’s upper bound tightens because both N successes sit in 2 of the 10 repo clusters.

WF versus F is indistinguishable: one proof apart, exact McNemar 4/3 p=1.0, repo-clustered difference +0.58pp [+0.0, +1.85] with 34% of resamples at or below zero — and WF routed 94% of its tool calls to formal search anyway, so SorryDB says essentially nothing about the Brain. The Brain-only arm W, present in every other phase, was never run here — a design gap, and precisely the arm that would have made the phase informative about the package contrast. The proving phase is therefore uninformative about the Brain, not evidence of no effect. Tools-versus-none is descriptive but consistent: the no-tools arm burns its budget (70 of its 168 rows produce nothing), while tool arms raise proving four-to-five-fold and double honest abstention — with only 10 repo clusters we leave it descriptive.

The benchmark itself decayed: the original freeze lost 74 of 164 tasks (45%) in about six months to rewritten upstream history — pinned commits force-pushed or rebased out of existence. The rot was invisible to cheap checks, because GitHub’s archive endpoint answers a preflight HEAD request with 200 for dead commits; only actually fetching every pin reveals it (full narrative and bookkeeping: Supplement §S7, Supplement §S9).

7 Limitations and discussion

Statistical scope. One model per phase, one seed per (task, arm), no run-to-run variance estimate anywhere. n=100 tasks in 44 commit clusters is underpowered for between-package contrasts — the D–E interval spans −0.09 to +0.28 — and nothing in this paper ranks tool packages in the join’s favor; the one decisive between-package contrast (F over W on premise retrieval, §5) runs the other way and is exploratory. The ProofNet# eval split was not re-typechecked under the repaired oracle: 280 raw-flagged, repaired-clean rows (A 39, B 45, C 68, D 47, E 81) are hard per-arm upper bounds on the gains there, and the instrument bias direction (against C/E) applies to that table too.

Endpoint validity. The preregistered semantic endpoint has still only been graded by an uncalibrated LLM judge that shares a vendor with the subject models; the human calibration and the 36-task discordant queue are pending. Grounded typecheck is a grounding measure, not a faithfulness measure, and §4.2 shows the two can rank arms oppositely. The repaired hallucination oracle’s flagged class is only 43–57% precise out of sample (recall 3/3; §4.1, Supplement §S3), so all per-arm hallucination rates are upper bounds on true rates.

Attribution. D bundles the join, the verifier, curated metadata, and a distinct interface; the E repair ran later than D, so a time confound remains on that contrast; E failed its manipulation check. Attribution of D’s tools-versus-none effect between join and verifier awaits the 2×2 factorial.

A warning for the field. The contamination-by-endpoint interaction (§4.2) generalizes: a contaminated task set does not merely inflate scores, it can *invert rankings* when one endpoint rewards transcription and another rewards grounded generation. Post-cutoff sets decay, and a holdout claim must name every index it is held out from — ours missed two (the mutable checkout; our own later discovery-fold channel).

Conflict and AI-use disclosure. The author built and operates the evaluated system. The study — runs, graders, and the corrective analyses in this revision, including the blinded protocols — was executed by AI agents (Claude Code) under the author’s direction, with human review of design and conclusions. The judge model and subject models are from the same vendor as the executing agents.

8 Conclusion

What this study establishes: giving a Lean agent formal-tool access lifts grounded typecheck output well above the no-tools floor (+16 to +27 points; clustered $p=0.0004$ for the Brain arm) and collapses flagged citation hallucination (6–11% of runs versus 30–37%, upper bounds under the repaired oracle); the sharpest mechanism signal is the existence verifier’s generate-then-verify loop, whose value survives every correction and converges with independent work (DDR). What it does not establish: that the curated join beats the same corpora unjoined — on grounded typecheck, judged equivalence, retrieval, or kernel-verified proving, every between-package contrast in the join’s favor is inconclusive once instruments are repaired and clustering is respected; the one decisive retrieval contrast (premise retrieval, $F - W +0.27$, $p=2.3e-6$) favors plain formal search; the bare-union ablation shows the union itself inert and the WF gain manual-driven; and the join’s retrieval value measured here is verification and routing rather than curated content. The decisive next experiment is the 2×2 factorial $\{\text{join, no join}\} \times \{\text{verifier, none}\}$ on a frozen post-cutoff set with human-calibrated equivalence grading and frozen-snapshot search infrastructure.

References

- CRAMF — Lu et al., *Automated Formalization via Conceptual Retrieval-Augmented LLMs*, arXiv:2508.06931 (preprint).
- Aria — Wang et al., *Aria: An Agent for Retrieval and Iterative Auto-Formalization via Dependency Graph*, arXiv:2510.04520 (ICLR 2026 poster; numbers quoted from v2, 2026-07-02).
- DRIFT — Zhang et al., *DRIFT: Decompose, Retrieve, Illustrate, then Formalize Theorems*, arXiv:2510.10815 (ICLR 2026).
- DDR — Wang et al., *Improving Autoformalization Using Direct Dependency Retrieval*, arXiv:2511.11990 (preprint).
- TheoremGraph — Kurgan et al., *TheoremGraph: Bridging Formal and Informal Mathematics*, arXiv:2606.25363 (preprint).
- LeanSearch-v2 — Gao et al., *LeanSearch v2: Global Premise Retrieval for Lean 4 Theorem Proving*, arXiv:2605.13137 (preprint; also the source of the MathlibQR/MPR data and the published anchor rows).
- ProofNetVerif / BEq+ / ProofNet# — Poiroux et al., *Improving Autoformalization using Type Checking*, arXiv:2406.07222.
- SorryDB — arXiv:2603.02668, sorrydb.org.